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Testing Heavy Neutral Leptons with Multi-Messenger Observations of NGC 1068

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The active galactic nucleus NGC 1068, recently identified as a high-energy neutrino source by IceCube, exhibits a GeV γ-ray flux that exceeds predictions from opaque corona cascade models. This tension suggests the existence of non-attenuated energy transport mechanisms from the compact core. We investigate whether heavy neutral leptons (HNLs), coupled to the Standard Model through mixing with muon neutrinos, can provide such a channel, and we derive the resulting *exclusionary constraint* on the HNL parameter space from the multi-messenger observations of NGC 1068. We calculate the production of HNLs in charged-pion and muon decays associated with hadronic interactions, explicitly tracking the mass-dependent polarization redistribution. We then derive the spectrum from the visible $N \to \nu_\mu + e^+ + e^-$ decay channel and evaluate its secondary electromagnetic emission. We find that the predicted photon flux is at least 7 to 8 orders of magnitude below the observational upper limits once experimental bounds on $|U_{\mu N}|^2$ are applied, which decisively excludes this HNL channel as the origin of the γ-ray excess. Even under an optimistic estimate in which particle transportation is neglected, the conclusion remains unchanged, and the exclusion of this channel is correspondingly strengthened. Nevertheless, the polarization-resolved framework developed here remains applicable to broader studies.

Keywords: HNL; Inverse Seesaw; multi-messenger; NGC 1068

1. Introduction

Multi-messenger astronomy has turned astrophysical accelerators into laboratories for physics beyond the Standard Model (BSM). The IceCube collaboration has identified the active galactic nucleus (AGN) NGC 1068 as a point source of neutrinos at energies around 1 TeV,[1] and the KM3NeT/ARCA telescope has reported a high-energy event candidate possibly reaching 220 PeV.[2] These observations establish that hadronic acceleration operates in at least some AGN cores, and they fix the neutrino flux that any transport mechanism in such a source must reproduce.

The same source is anomalous in γ-rays. The GeV γ-ray flux of NGC 1068

measured by Fermi-LAT exceeds the predictions of opaque corona cascade models, in which the γ rays are generated in the optically thick core and attenuated before escape.[3,4] The excess therefore points to a transport channel that carries energy out of the compact core without being absorbed, and that converts this energy into observable photons only beyond the opaque region. Neutral, weakly interacting particles are natural candidates for such a channel: they escape the core freely and decay or scatter in the surrounding environment, where their secondaries radiate. The question we address is whether a specific BSM particle can play this role.

HNLs are motivated here on two levels: theoretically, they are among the simplest BSM particles that generate neutrino mass; phenomenologically, an MeV-scale HNL mixed with the muon neutrino is produced in the dominant neutrino production chain of NGC 1068 (pion and muon decays) and possesses a visible decay channel, so that it is a qualified candidate for the transport channel of this source. The non-zero but extremely small active neutrino masses require physics beyond the Standard Model,[5,6] and HNLs are among the simplest mediators of neutrino mass, simultaneously providing a dark-matter candidate and a target for laboratory searches.[7] In the Inverse Seesaw (ISS) mechanism the smallness of the light neutrino mass is controlled by a small lepton-number-violating parameter rather than by a very large Majorana mass, so that the mixing with active neutrinos can remain large enough to be phenomenologically relevant. Terrestrial experiments have not yet observed a definite HNL signal, but they constrain the mixing with the muon neutrino to $|U_{\mu N}|^2 \lesssim 10^{-4}$ for $3\,\mathrm{MeV} < m_N < 19.5\,\mathrm{MeV}$;[8] the bound is slightly weaker but of the same order over the 10 MeV to 30 MeV range considered here. These theoretical and phenomenological conditions are both met in this source, which is why we single out HNLs rather than an arbitrary BSM particle; other neutral candidates, such as axion-like particles, have been studied separately for NGC 1068,[9] and we do not attempt a comparative survey here.

This motivates a dedicated calculation. A precise statement cannot rely on the massless neutrino results: for m_N in the 10 MeV to 30 MeV range the HNL is non-relativistic in the parent rest frame, phase space is strongly suppressed, and the helicity produced in the weak decay is redistributed by the boost in a mass-dependent way. This is visible already at the production stage: as shown in Figs. (2)–(4) below, right-handed HNLs, which are absent in the massless limit, become an appreciable fraction of the pion-decay yield as m_N approaches the phase-space boundary, and Fig. (4) shows the same effect in muon decay. These polarization states are not washed out in the subsequent decay: the visible channel $N \to \nu_\mu + e^+ + e^-$ inherits the helicity through the angular distributions $g_e^{(i)}(x, \cos\theta)$ of Eqs. (24)–(25), so the electron spectrum, and hence the γ-ray yield, depends on the polarization of the parent HNL. A polarization-resolved treatment is therefore adopted here to convert the NGC 1068 multi-messenger data into a quantitative statement about this channel, and it is also what makes the framework reusable for other sources and other HNL mass windows.

In this paper we test whether this HNL channel can account for the γ-ray excess of NGC 1068, and we derive the corresponding *exclusionary constraint* on the HNL parameter space. We compute the HNL source spectrum from $\pi^{\pm} \to \mu^{\pm} + N$ and $\mu^{\pm} \to N + e^{\pm} + \bar{\nu}_e$, separating the two helicity states and boosting to the laboratory frame while tracking the polarization redistribution. We then derive the electron/positron spectrum from the visible decay $N \to \nu_\mu + e^+ + e^-$, propagate these leptons in a representative environment of NGC 1068, and compute the resulting electromagnetic emission with GAMERA.[10] Comparing with the observed γ-ray flux, we find that the predicted photon flux lies at least 7 to 8 orders of magnitude below the observational upper limits once the experimental bound on $|U_{\mu N}|^2$ is applied. Even under an optimistic estimate in which particle transport is neglected, the conclusion is unchanged. The HNL channel is therefore decisively excluded as the origin of the γ-ray excess, and the exclusionary constraint on this channel is correspondingly strong.

The paper is organized as follows. Section 2 gives the HNL coupling, the ISS parameter range, the single-Dirac approximation, and the decay channels. Section 3 presents the polarization-resolved calculations of HNL production in pion and muon decays and of the visible three-body decay $N \to \nu_\mu + e^+ + e^-$, and collects them into the electron source function injected into the radiation calculation. Section 4 applies this source function to the environment of NGC 1068, compares with observations, and gives the suppression budget and its robustness. Section 5 summarizes our conclusions.

2. ISS Model: Parameters and Prerequisites

This section provides the four ingredients needed below: the coupling of HNLs to the SM, the allowed ranges of m_N and $|U_{\mu N}|^2$ in the ISS, the justification of the single-Dirac approximation, and the visible decay channel with its branching ratio. The production and decay calculations of Sec. 3 and the experimental bound and suppression budget of Sec. 4 rest on them.

2.1. *Coupling*

HNLs interact with the SM only through their mixing with the active neutrinos. Therefore any process that produces ν_μ also produces N, with a rate multiplied by $|U_{\mu N}|^2$. Section 3 uses this to compute N production in charged-pion and muon decay chains.

2.2. *ISS and parameter ranges*

In the Type-I seesaw:[11–16]

$$m_\nu = \frac{m_D^2}{M_R},\ m_N = M_R,\ |U|^2 = \left(\frac{m_D}{M_R}\right)^2 \tag{1}$$

For an observable M_R the mixing is usually too small to be probed. The Inverse Seesaw (ISS)[11–16] introduces another fermion field S, and the mass matrix reads

$$\left(\overline{\nu_L}\ \overline{\nu_R^c}\ \overline{S^c}\right)\begin{pmatrix} 0 & m_D & 0 \\ m_D & 0 & M \\ 0 & M & \mu \end{pmatrix}\begin{pmatrix} \nu_L^c \\ \nu_R \\ S \end{pmatrix} \tag{2}$$

where μ is a small lepton-number-violating mass term. For $\mu \ll M$ and $\mu \ll m_D$,

$$m_\nu = \mu\frac{m_D^2}{M^2},\ m_N = M,\ |U|^2 = \frac{m_\nu}{\mu} \tag{3}$$

The key point is that the smallness of the light neutrino mass is controlled by the small parameter μ rather than by a very large M, so that for comparable neutrino masses the mixing can be larger than in the Type-I case.

For the parameters used here, $|U|^2$ can reach the 10^{-4} level: taking $m_\nu \sim 0.05$ eV and $\mu \sim$ keV gives $|U|^2 = m_\nu/\mu \sim 10^{-4}$, while $\mu \ll m_D, M$ is satisfied. This is the same order as the experimental bound quoted in Sec. 1, and it is what allows Sec. 3 to take $|U|^2$ not small and Sec. 4 to use the bound $|U_{\mu N}|^2 \lesssim 10^{-4}$.

2.3. *Single-Dirac approximation*

For small μ the heavy states form a nearly degenerate pseudo-Dirac pair.[17,18] Over the range $m_N \simeq 10\,\text{MeV}$ to $30\,\text{MeV}$ and $\mu \ll m_D, M$ considered here, the mass splitting is much smaller than m_N, so that production and decay cannot resolve the two states; summing over helicities is therefore equivalent to a single Dirac fermion. The helicity-resolved calculation of Sec. 3, with $s = \pm 1$, rests on this approximation.

2.4. *Decay channels and visible branching ratio*

Because the HNLs considered here are lighter than the charged-pion decay threshold, their production and decay are governed by low-energy weak processes. They decay only into leptons: a three-neutrino channel (invisible) and a $\nu_\mu e^+ e^-$ channel (visible). Section 3.3 gives the explicit value $BR \simeq 0.11$, which is the first entry of the suppression budget in Sec. 4.3 and one of the three sources discussed in Sec. 5.

3. Polarization-Resolved HNL Production and Decay

HNLs interact only through their mixing with SM neutrinos, which implies that processes producing SM neutrinos will also produce HNLs, provided they are kinematically allowed. Under AGN conditions, high-energy cosmic rays undergo $p-p$ and $p-\gamma$ interactions, which are thought to be the dominant sources of secondary particles;[19] those processes create a large number of pions, whose decays subsequently produce high-energy cosmic rays, including neutrinos and γ-rays.[20] We consider the two production channels most directly related to the muon-neutrino sector, charged pion decay and muon decay, and then the visible decay of the HNL; polarization is tracked through all three stages.

Throughout this section, E_N and E denote laboratory-frame energies of the HNL and of the electron, respectively, while x and y are the corresponding dimensionless variables in the parent rest frame, defined where they first appear. The helicity label $s = \pm 1$ denotes the HNL spin projection along its momentum, with $s = +1$ for a right-handed HNL; the same convention is used in all three stages below.

3.1. *Production in pion decay*

The decay $\pi^+ \to \mu^+ + N$ is the HNL analogue of the process $\pi^+ \to \mu^+ + \nu_\mu$.[21]

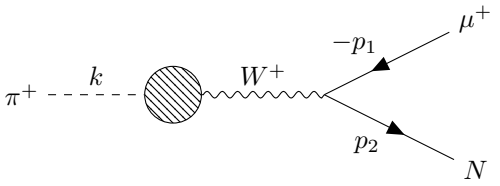


Fig. 1. Feynman diagram of π^+ decaying into μ^+ and massive HNLs.

By slightly modifying the process of SM pion decay, we find that the decay width to HNL is suppressed by the mixing angle and modified by a kinematic factor:

$$\frac{\Gamma_{\pi\to\mu N}}{\Gamma_{\pi\to\mu\nu}} = |U_{\mu N}|^2 \times \rho_\pi(m_N) \tag{4}$$

where

$$\rho_\pi(m_N) = \frac{(m_\pi^2(m_\mu^2 + m_N^2) - (m_\mu^2 - m_N^2)^2)\sqrt{\lambda(m_\pi^2, m_\mu^2, m_N^2)}}{m_\mu^2(m_\pi^2 - m_\mu^2)^2} \tag{5}$$

$$\lambda(x, y, z) = x^2 + y^2 + z^2 - 2xy - 2yz - 2zx \tag{6}$$

The function ρ_π encodes the phase-space suppression.

It is useful to determine the polarization of the HNLs, since it will play a role in the angular dependence when working on the decay of HNL. In the pion rest frame we have

$$\Gamma_{(s)} \sim (m_\mu^2 + m_N^2)m_\pi^2 - (m_\mu^2 - m_N^2)^2 - s(m_\mu^2 - m_N^2)\sqrt{\lambda(m_\pi^2, m_\mu^2, m_N^2)} \tag{7}$$

where s is the polarization of HNL, and $s = +1$ for right-handed polarized HNLs.

Since the parent pion is a spin-zero scalar, the decay is isotropic in its rest frame. However, the polarization of the massive HNL needs to be carefully tracked, as the subsequent Lorentz boost to the laboratory frame induces a mass-dependent polarization redistribution. For a pion spectrum $J(E_\pi)$, we get

$$Q_N^{(s)}(E_N) = \int_{\lambda^-}^{\lambda^+} J(\frac{E_N}{x})|U_{\mu N}|^2 \rho_\pi \times r_{(s)}(x, E_N) \times \frac{dx}{(\lambda^+ - \lambda^-) \cdot x} \tag{8}$$

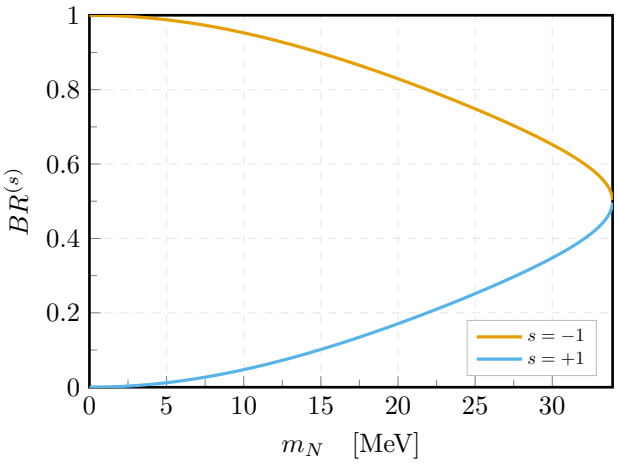


Fig. 2. Eq. (7) for different s. In pion decay, right-handed HNLs are more likely to be produced close to the phase space boundary.

where

$$\lambda^{\pm} = \frac{m_\pi^2 - m_\mu^2 + m_N^2 \pm \sqrt{\lambda(m_\pi^2, m_\mu^2, m_N^2)}}{2m_\pi^2} \tag{9}$$

$$\cos\theta_r = \frac{x\,(m_\pi^2 - m_\mu^2 + m_N^2) - 2m_N^2}{x\sqrt{(1 - \frac{m_N^2}{E_N^2})\,\lambda(m_\pi^2, m_\mu^2, m_N^2)}} \tag{10}$$

$$r_{(s)}(x, E_N) = \frac{\Gamma_{(s)}\frac{1+\cos\theta_r}{2} + \Gamma_{(-s)}\frac{1-\cos\theta_r}{2}}{\Gamma_{(s)} + \Gamma_{(-s)}} \tag{11}$$

Here we assume $E_\pi \gg m_\pi$ to simplify the boost process. This resembles the SM neutrino case,[20] with modifications arising from HNL mass and helicity separation.

3.2. *Production in muon decay*

Muons can decay into HNLs via a process analogous to their standard decay into ν_μ. We find the decay width of $\mu^- \to Ne^-\bar{\nu}_e$ has the same form as the previously studied τ decay process,[22] since this process is symmetric between μ and τ, which is

$$\frac{\Gamma_{\mu\to N}}{\Gamma_{\mu\to\nu_\mu}} = |U_{\mu N}|^2 \times \rho_\mu(m_N) \tag{12}$$

where

$$\rho_\mu(m_N) = -r_N^8 + 8r_N^6 - 24r_N^4\ln(r_N) - 8r_N^2 + 1, \quad r_N = m_N/m_\mu \tag{13}$$

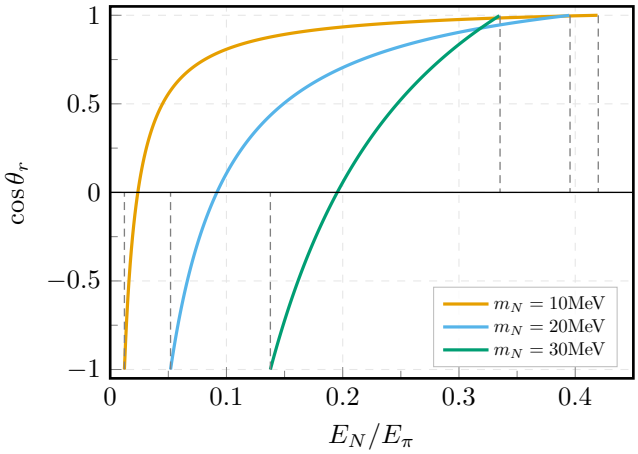


Fig. 3. Eq. (10) visualized with different m_N values. Eq. (10) determines how much helicity flip will happen during Lorentz boost. When m_N is small $\cos\theta_r \approx 1$, combined with Fig. (2), HNLs produced are almost left-handed; as m_N approaches the edge of phase space, the helicity flip is more significant and right-handed HNLs cannot be ignored.

The helicity-dependent spectrum of the HNL in the muon rest frame is given by

$$\frac{d\Gamma}{dE} = \frac{G_F^2}{24\pi^3}|U_{\mu N}|^2 \times |\mathbf{p}| \cdot [3(m_\mu^2 + m_N^2)E - 4m_\mu E^2 - 2m_N^2 m_\mu - s \cdot |\mathbf{p}|(m_N^2 + 3m_\mu^2 - 4m_\mu E)] \tag{14}$$

Several examples are plotted in Fig. (4).

A fully spin-correlated treatment could retain the polarization of the parent muons and the angular relation between their spin and decay products. Here we neglect parent-muon polarization as a simplifying approximation. This approximation isolates the HNL mass and polarization dependence but should not be interpreted as a demonstrated consequence of complete magnetic depolarization in the source. For a given muon spectrum, we have

$$Q_N^{(s)}(E_N) = \int_{\frac{m_N^2}{m_\mu^2}}^1 \frac{dy}{2y}\, Q_\mu(\frac{E_N}{y})|U_{\mu N}|^2 \rho_\mu \cdot \int_{x^-}^{x^+} \frac{dx}{\sqrt{x^2 - \frac{m_N^2}{m_\mu^2}}} \cdot \left(\frac{1+\cos\theta_r}{2} g_N^{(s)}(x) + \frac{1-\cos\theta_r}{2} g_N^{(-s)}(x)\right) \tag{15}$$

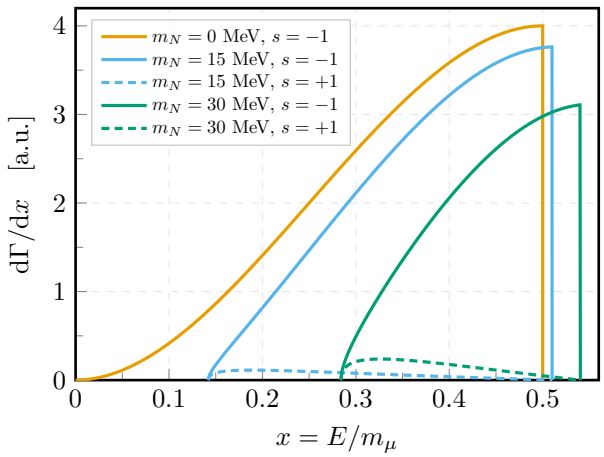


Fig. 4. Energy spectrum of HNL when muons decay into HNLs in muon rest frame. Normalized under $m_N = 0$, which basically resembles SM neutrino, with no right-handed product produced. When m_N increases, the shape of the spectrum remains roughly the same, but the maximum energy increases while the total decay width decreases, yielding a non-negligible fraction of right-handed HNLs.

where

$$x^+ = \frac{m_N^2 + m_\mu^2}{2m_\mu^2},\ x^- = \frac{1}{2}\left(y + \frac{1}{y}\cdot\frac{m_N^2}{m_\mu^2}\right) \tag{16}$$

$$\cos\theta_0 = \frac{y - x}{\sqrt{x^2 - \frac{m_N^2}{m_\mu^2}}} \tag{17}$$

$$\cos\theta_r = \frac{1}{y\sqrt{1 - \frac{m_N^2}{E_N^2}}}\left(\sqrt{x^2 - \frac{m_N^2}{m_\mu^2}} + x\cos\theta_0\right) \tag{18}$$

where $g_N^{(s)}(x)$ is normalized as

$$\int_{\frac{m_N}{m_\mu}}^{x^+} dx \sum_{s=\pm 1} g_N^{(s)}(x) = 1 \tag{19}$$

3.3. *Visible three-body decay*

Kinematically, low-mass HNLs in the ISS model will exclusively decay into other leptons. Possible products are $\nu_\mu e^- e^+$ and $\nu_\mu \nu_l \bar{\nu}_l$. These processes have been well studied;[22–24] their decay widths are given by

$$\Gamma_{N\to 3\nu} = \frac{G_F^2 m_N^5}{192\pi^3} \times |U_{\mu N}|^2 \tag{20}$$

$$\Gamma_{N\to \nu e\bar{e}} = \frac{G_F^2 m_N^5}{192\pi^3} \times |U_{\mu N}|^2 \times C_1^e \tag{21}$$

where

$$C_1^e = \frac{1}{4}\left(1 - 4\sin^2\theta_W + 8\sin^4\theta_W\right) \tag{22}$$

To simplify the problem, we neglect the electron mass. The 3ν final state includes all neutrino flavor combinations.

An HNL produced in the central region can affect the electromagnetic signal only if its energy is ultimately transferred to visible charged particles, thus the branching ratio for an HNL to undergo a visible decay is

$$BR_{N\to\nu e\bar{e}} = \frac{C_1^e}{1 + C_1^e} \approx 0.11 \tag{23}$$

This results in substantial energy loss, as only approximately one-tenth of the HNL decays are observable.

The angular and energy distributions of electrons/positrons are

$$\begin{aligned} g_e^{(1)}(x,\cos\theta) =& \frac{8}{C_1^e}\cdot x^2\cdot\{[(4-16s_W^2+64s_W^4)x-(1-4s_W^2+28s_W^4)]\cos\theta \\ & -(4-16s_W^2+64s_W^4)x+(3-12s_W^2+36s_W^4)\} \end{aligned} \tag{24}$$

$$\begin{aligned} g_e^{(2)}(x,\cos\theta) =& \frac{16}{C_1^e}\cdot x^2\cdot\{[(6-24s_W^2+32s_W^4)x-(3-12s_W^2+14s_W^4)]\cos\theta \\ & -(6-24s_W^2+32s_W^4)x+(3-12s_W^2+18s_W^4)\} \end{aligned} \tag{25}$$

Here, $g_e^{(1)}$ and $g_e^{(2)}$ represent the differential distributions for the outgoing electron (e^-) and positron (e^+), respectively, with $x = E_e/m_N$ and $s_W = \sin\theta_W$. The angle θ is defined between the outgoing lepton momentum and the parent HNL spin. To obtain the total injected lepton spectrum, we sum over both e^- and e^+:

$$\begin{aligned} Q_e^{(Lab)}(E) = BR_{N\to\nu e\bar{e}}\cdot\int_E^{+\infty} dE_N\, Q_N^{(s)}(E_N) \\ \cdot\int_{\frac{E}{2E_N}}^{\frac{1}{2}}\frac{dx}{xE_N}\sum_{i=1,2} g_e^{(i)}(x, s(\frac{E}{xE_N}-1)) \end{aligned} \tag{26}$$

Evaluating the angular and energy integrals yields

$$\begin{aligned} Q_e^{(Lab)}(E) =& BR_{N\to\nu e\bar{e}} \\ & \cdot\int_0^1\frac{dy}{3y}\,Q_N^{(s)}(\frac{E}{y})[16y^3-27y^2+11-s\cdot(32y^3-63y^2+36y-5)] \end{aligned} \tag{27}$$

where $y = E/E_N$. Notice that the dependence on the Weinberg angle θ_W cancels completely in the sum. In the massless limit $m_e \to 0$, helicity-flipping mass terms vanish, and the vector–axial-vector (V-A) interference terms are odd under charge conjugation ($e^- \leftrightarrow e^+$); they therefore cancel when summing over the electron and positron yields, leaving the total spectral shape independent of $\sin^2\theta_W$.

4. HNL in NGC 1068

According to IceCube observations, NGC 1068 has been identified as a neutrino point source, and the dominant channel of neutrino production is charged pion decay.[4] By combining the analytical neutrino spectrum from pion decays[20] with theoretical models of NGC 1068,[4] we adopt the following parameterization as an approximation of the pion spectrum:

$$\Phi_{\pi^\pm}(E) = \Phi_0 \cdot \left(\frac{E}{1\,\mathrm{GeV}}\right)^{-\Gamma} \cdot e^{-(E/E_c)^\beta} \tag{28}$$

with best-fit values

$$\Phi_0 = 4.7 \times 10^{-12}\,\mathrm{GeV}^{-1}\,\mathrm{cm}^{-2}\,\mathrm{s}^{-1}, \Gamma = 0.1, E_c = 6.2\,\mathrm{GeV}, \beta = 0.3 \tag{29}$$

visualized in Fig. (5). These parameters are obtained by fitting this form to the neutrino flux of the theoretical model,[4] as the uncertainty and narrow energy band of observed flux of IceCube makes it harder to produce a whole picture for parameters fitting. The spectrum provides the parent-particle input for the production calculation of Sec. 3.

4.1. *Decay length and the escape assumption*

For HNLs to mediate energy transport from the opaque core to a transparent external environment, their decay length must be sufficiently large. We have

$$l_{dec} = \frac{E_N}{m_N}\, c\, \frac{\hbar}{\Gamma_{tot}} = \frac{192\pi^3 c\hbar}{G_F^2(1+C_1^e)} \cdot \frac{E_N}{m_N^6 |U_{\mu N}|^2} \tag{30}$$

For demonstration, setting $|U_{\mu N}|^2 = 10^{-5}$, $m_N = 20\,\mathrm{MeV}$, and $E_N = 1\,\mathrm{TeV}$ yields $l_{dec} \approx 0.4\,\mathrm{pc}$, which implies that HNLs are unlikely to escape into interstellar space, while some decays will still occur inside the optically opaque region.

Because $l_{dec} \propto E_N/(m_N^6|U_{\mu N}|^2)$, the decay length is extremely sensitive to m_N: over the 10 MeV to 30 MeV range it varies by more than two orders of magnitude. Since $l_{dec} \propto 1/|U_{\mu N}|^2$, at the experimental bound $|U_{\mu N}|^2 \lesssim 10^{-4}$ (the representative bound over the 10 MeV to 30 MeV range) the mixing is one order of magnitude larger than the value used above, so that l_{dec} is correspondingly shorter by about one order of magnitude (approximately 0.04 pc). Most HNLs then decay inside the opaque core and their visible products are absorbed before escape. This further suppresses the signal and strengthens the exclusion. We therefore adopt the optimistic assumption that all HNLs decay beyond the core and that their energy escapes unattenuated; this assumption maximizes the predicted flux and makes the resulting exclusion conservative.

4.2. *Environment and radiation calculation*

We refer to the parameters of the AGN torus model.[25] For our approximation we set the magnetic field strength to 139 mG, dust temperature to 1500 K, ambient density

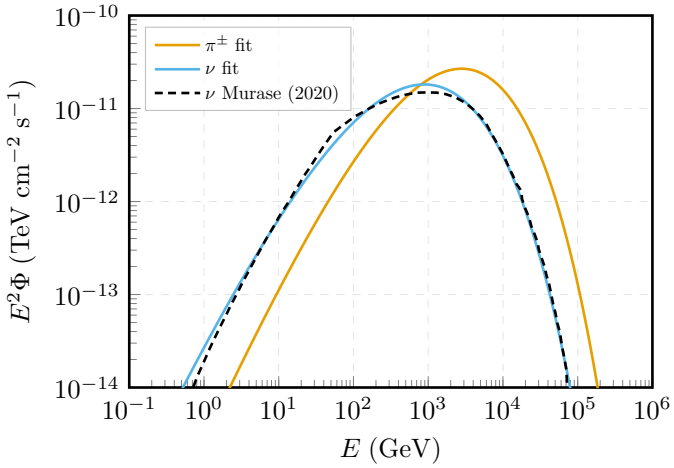


Fig. 5. Parametrization of the pion spectrum in NGC 1068 (Eq. (28)) for theoretical prediction.[4]

to $2.30 \times 10^5\,\mathrm{cm}^{-3}$, and bolometric luminosity to $9.55 \times 10^{44}\,\mathrm{erg\,s}^{-1}$, and assume that electron-photon conversion happens at about 0.4 pc away from the AGN core, corresponding to the inner edge of the torus in this model.

The electron/positron source function $Q_e^{(Lab)}(E)$ derived in Sec. 3.3 is injected into GAMERA,[10] which computes synchrotron, inverse-Compton (on the dust and CMB photon fields), and bremsstrahlung emission from these leptons in the adopted environment. Internal $\gamma\gamma$ absorption is neglected as an optimistic assumption; including it would further reduce the predicted flux. The resulting photon spectrum is shown in Fig. (7).

4.3. *Comparison with observations and the suppression budget*

Factoring out the mixing angle, i.e. setting $|U_{\mu N}|^2 = 1$, the predicted photon flux lies three to four orders of magnitude below the observed excess. This residual suppression arises mainly from two results already obtained above: first, the small visible branching ratio $BR \simeq 0.11$, which contributes about one order of magnitude; second, the fact that the HNL carries only a fraction of the parent pion/muon energy into the visible channel, together with the radiation efficiency in the adopted environment, which together account for the remaining orders. The phase-space factors ρ_π, ρ_μ are of order unity over most of the 10 MeV to 30 MeV range and become significant only near the kinematic endpoint, so they are not counted as a source of orders of magnitude. Applying the experimental bound $|U_{\mu N}|^2 \lesssim 10^{-4}$ (the representative value over this range) adds four more orders of magnitude, bringing the total suppression to seven to eight orders. Even if the mixing angle were set to unity, the channel would remain three to four orders of magnitude short of the data.

Two qualifications limit the scope of this numerical claim but do not motivate

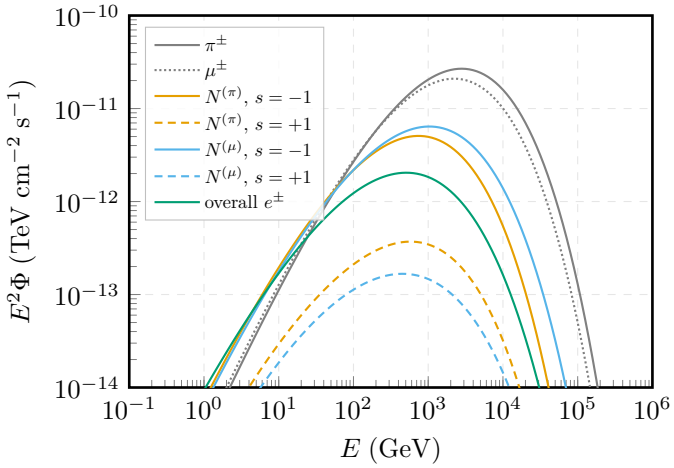


Fig. 6. HNL production source function from pion and muon decay for NGC 1068, using Eq. (28) as source pion spectrum. The polarization of muon is omitted. HNL mass is set to $m_N = 10\,\mathrm{MeV}$, and all HNL spectrums are scaled to $|U_{\mu N}|^2 = 1$ to better show the physics. For $m_N = 10\,\mathrm{MeV}$, right-handed neutrino flux is 1 to 2 orders of magnitude smaller than left-handed one.

treating the present calculation as evidence for a detectable HNL signal. First, the experimental bound on $|U_{\mu N}|^2$ is mass dependent, and the value quoted for a particular low-mass interval should not be extrapolated unchanged over the entire 10 MeV to 30 MeV range, although the difference is minor. Second, an accurate observable flux from a finite external region must account for the probability of decay within that region; the present treatment instead assumes that all HNLs deposit their visible products in the chosen zone, which makes the estimate optimistic rather than fully propagated. A more detailed discussion of these points is given in Sec. 5. The comparison in Fig. (7) uses the Fermi-LAT point-source upper limits.[3]

The constraint obtained here is *exclusionary*: in the range $m_N \simeq 10\,\mathrm{MeV}$ to $30\,\mathrm{MeV}$, even with the largest mixing allowed by experiment ($|U_{\mu N}|^2 \simeq 10^{-4}$), the predicted flux remains seven to eight orders of magnitude too low, so that the channel is excluded as an explanation of the γ-ray excess. This does not change the laboratory upper limit on $|U_{\mu N}|^2$, but it rules out the channel as an explanation over this mass range.

5. Conclusions

In this article we have detailed the calculation of the production and decay of HNLs mixed with the muon neutrino sector, and applied it to the multi-messenger data of NGC 1068. For $\pi^{\pm} \to N + \mu^{\pm}$ and $\mu^{\pm} \to N + e^{\pm} + \bar{\nu}_e$, we computed the HNL spectra separated by polarization, boosted them to the laboratory frame while tracking the mass-dependent helicity redistribution, and then derived the electron/positron spectrum from the visible decay $N \to \nu_\mu + e^+ + e^-$ for each initial

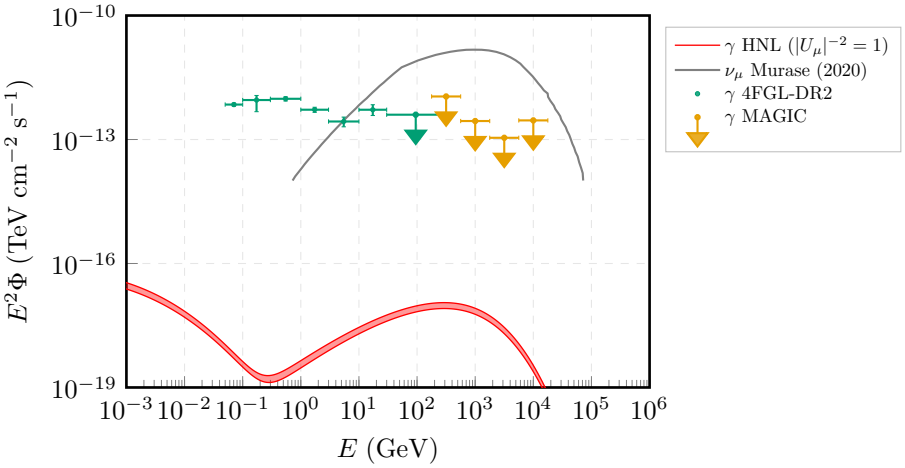


Fig. 7. Possible HNL generated photon spectrum compared with observations. m_N = 10 MeV to 30 MeV is shown as a range. Even ignoring the mixing angle, which is smaller than 10^{-4}, the flux is 3 to 4 orders of magnitude smaller than the observational upper limit, which means it is at least 7 to 8 orders of magnitude smaller and therefore can be completely ignored in observation.

polarization. Adopting a representative external radiation environment for NGC 1068, we obtained an HNL-associated γ-ray flux that is at least 7 to 8 orders of magnitude below the current observational limits once the experimental bound on $|U_{\mu N}|^2$ is applied. Even with the mixing angle set to unity, the predicted flux remains three to four orders of magnitude below the data. The weakly coupled HNL channel is therefore decisively excluded as the primary mechanism for the anomalous γ-ray excess of this source.

The robustness of this exclusion rests on two points, discussed qualitatively in Sec. 4.3 and made explicit here. First, the experimental bound on $|U_{\mu N}|^2$ is mass dependent; the value $|U_{\mu N}|^2 \lesssim 10^{-4}$ quoted for 3 MeV $< m_N <$ 19.5 MeV should not be extrapolated unchanged over the entire 10 MeV to 30 MeV range. Over this range the bound is slightly weaker, but the change is minor and does not affect the order-of-magnitude conclusion. Second, an accurate observable flux from a finite external region must account for the probability that an HNL decays within that region, which depends on its energy, mass, and mixing. The present calculation instead assumes that all relevant HNLs deposit their visible decay products in the chosen zone; this makes the estimate optimistic rather than a fully propagated flux prediction. Both qualifications limit the quantitative reach of the numbers quoted above, but neither reverses the conclusion: relaxing them can only reduce the predicted flux further, so the exclusion is conservative.

The exclusion itself is informative. It removes the simplest weakly coupled transport channel for the NGC 1068 γ-ray excess, thereby sharpening the case for alternative explanations, and it converts the multi-messenger data of this source into

an *exclusionary constraint* on the HNL parameter space that is independent of, and complementary to, laboratory bounds. The total suppression of seven to eight orders of magnitude decomposes into four orders from the mixing angle (at the experimental bound 10^{-4}) and three to four orders from $BR \simeq 0.11$ together with the HNL energy fraction and the radiation efficiency of the source environment (see Sec. 4.3). Even at the experimental upper bound on the mixing, BR together with the energy fraction and efficiency still suppress the signal; this does not change the laboratory upper limit on $|U_{\mu N}|^2$, but it rules out the channel as an explanation.

The anomaly therefore needs to be explained through a different process, most likely a refined model of the AGN environment on which current electromagnetic cascade calculations are based.[4] Alternatively, energy transport mediated by other weakly interacting particles remains a possibility. The polarization-resolved framework developed here is not specific to this conclusion: it applies to any source in which HNLs are produced in hadronic decay chains, and it can be reused for other HNL mass windows and for other weakly coupled neutral particles whose polarization affects their visible decay products. In this sense the negative result obtained here is accompanied by a controlled and reusable tool for future multi-messenger tests of BSM physics.

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